DEEP LEARNING SHOWCASE

Production Background Removal

High-Fidelity Portrait & Object Matting with ResNet-34 U-Net

Artifact-Free Bilinear Decoder • Soft BCEDice Combined Loss • Multi-GPU PyTorch Pipeline

98.39%
VAL DICE SCORE
Harmonic mean of precision & recall

96.92%
VAL IOU SCORE
Intersection over Union (Jaccard)

0.0502
FINAL LOSS
Down from 0.4878 initial loss

40 Epochs
COSINE LR SCHEDULE
Smooth decay with backbone warmup

PHASE 01

Pretrained Feature Extractor

ResNet-34 backbone pre-trained on ImageNet. Initial 2-epoch freeze warmup prevents disruptive gradient propagation into pretrained weights.

PHASE 02

Smooth Bilinear Decoder

Replaces standard transposed convolutions with bilinear interpolation + convolution, eliminating checkerboard boundary artifacts on hair and silhouette edges.

PHASE 03

Compound Soft Loss & Matting

Joint BCEDice loss directly penalizes boundary misalignments. Aspect-ratio preserving letterbox pipeline restores outputs up to 32+ Megapixels.

Training Dynamics & Convergence Performance

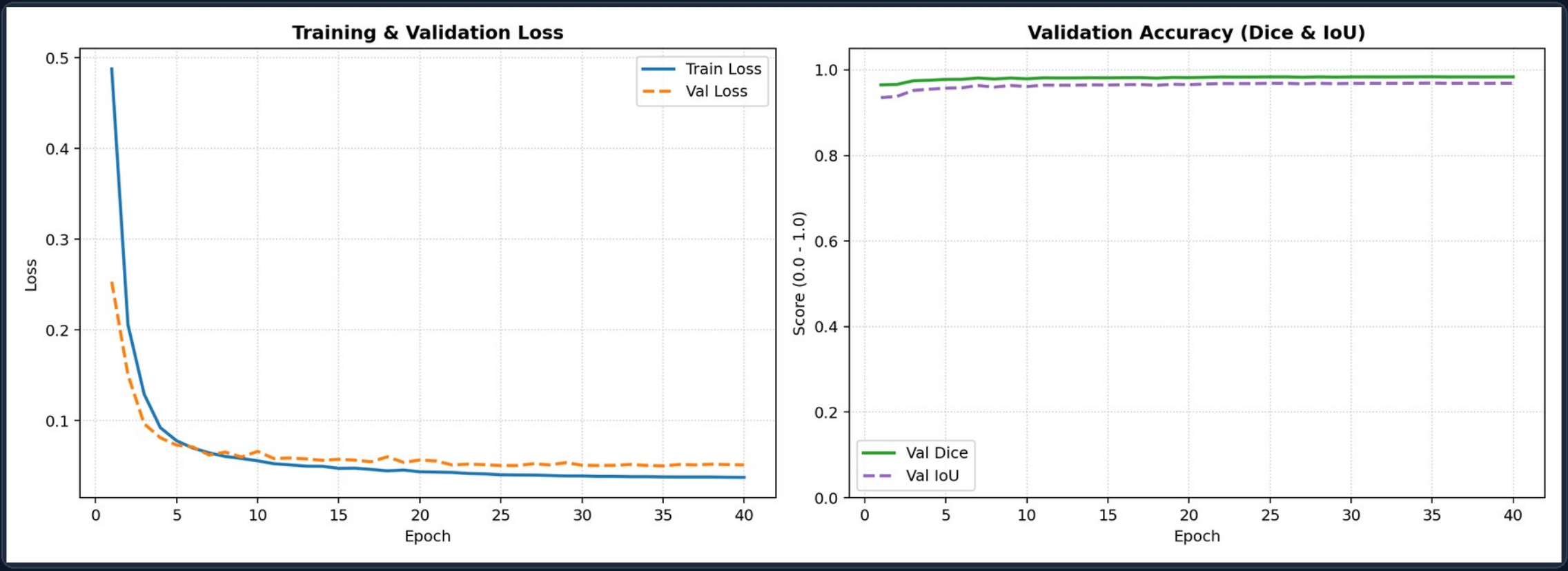
Quantitative Analysis: 40 Epochs with Cosine Annealing Learning Rate & Warmup Backbone Freezing

98.39%
PEAK VALIDATION DICE
Harmonic mean of precision & recall

96.92%
PEAK VALIDATION IOU
Jaccard index of segment overlap

0.0502
BEST VALIDATION LOSS
Combined Soft BCEDice Loss

40 Epochs
COSINE ANNEALING LR
Decayed from 1e-3 down to 1e-6



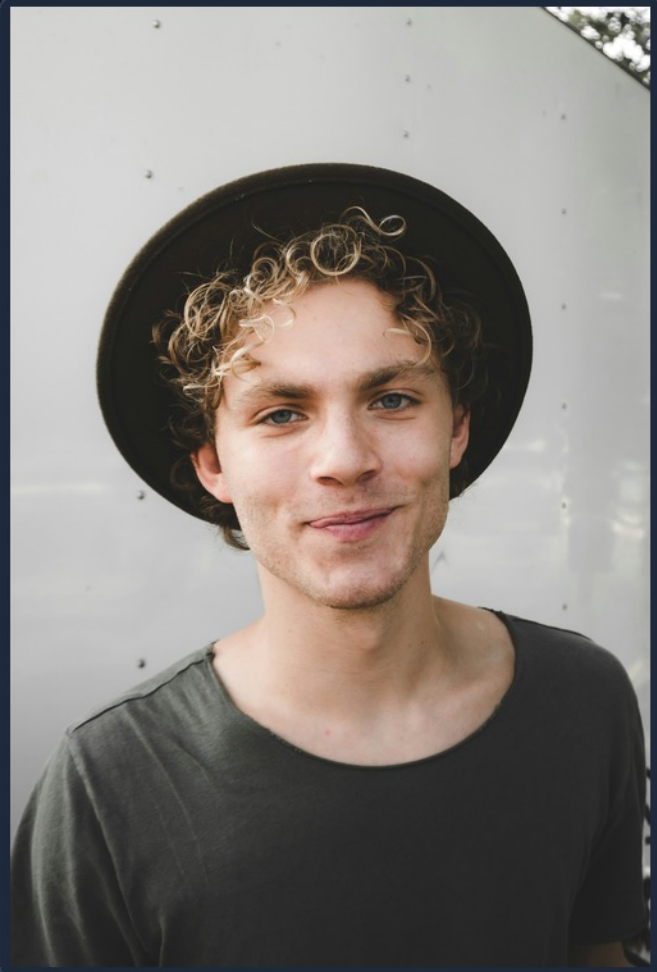
P3M-10k Dataset | ResNet-34 Feature Extractor | Bilinear Smooth Decoder

Visual Comparisons →

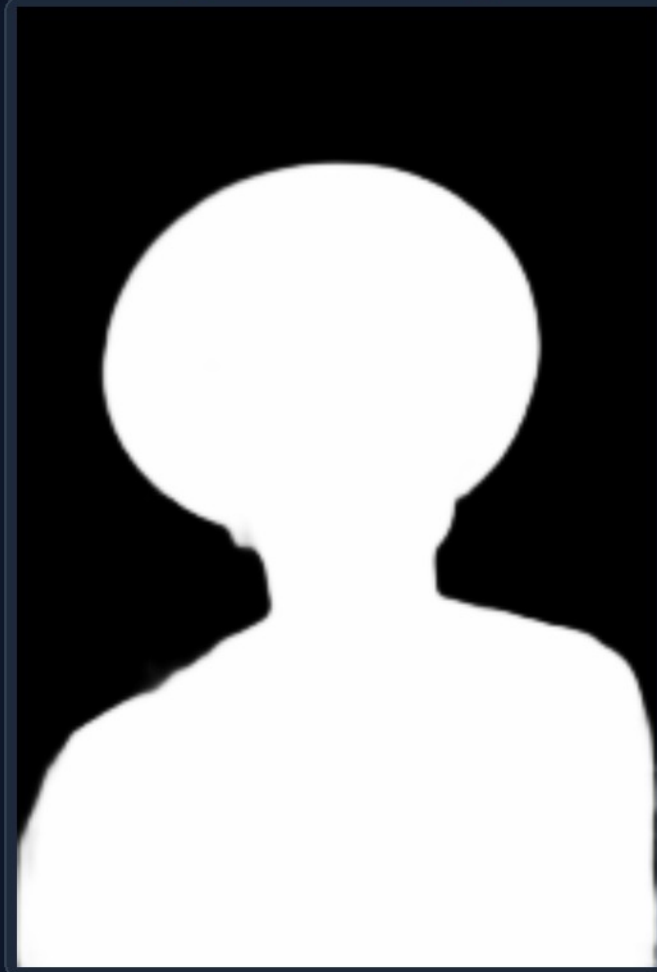
Case Study 01: Delicate Edge & Shoulder Silhouette Isolation

Native Input Resolution: 2662 × 3993 px | Deep Learning Architecture: ResNet-34 U-Net (Bilinear Decoder)

1. Original Image



2. Predicted Alpha Mask



3. Removed Background



Case Study 02: Studio Portrait Matting with Complex Attire

Native Input Resolution: 960 × 1280 px | Deep Learning Architecture: ResNet-34 U-Net (Bilinear Decoder)

1. Original Image



2. Predicted Alpha Mask



3. Removed Background



Case Study 03: Close-Up Portrait Hair Strand Preservation

Native Input Resolution: 896 × 1056 px | Deep Learning Architecture: ResNet-34 U-Net (Bilinear Decoder)

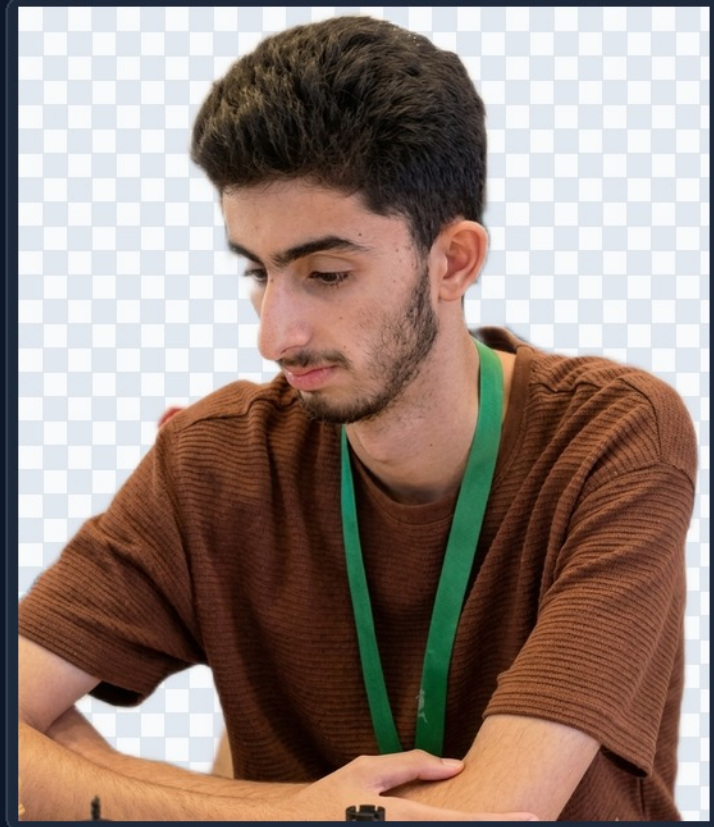
1. Original Image



2. Predicted Alpha Mask



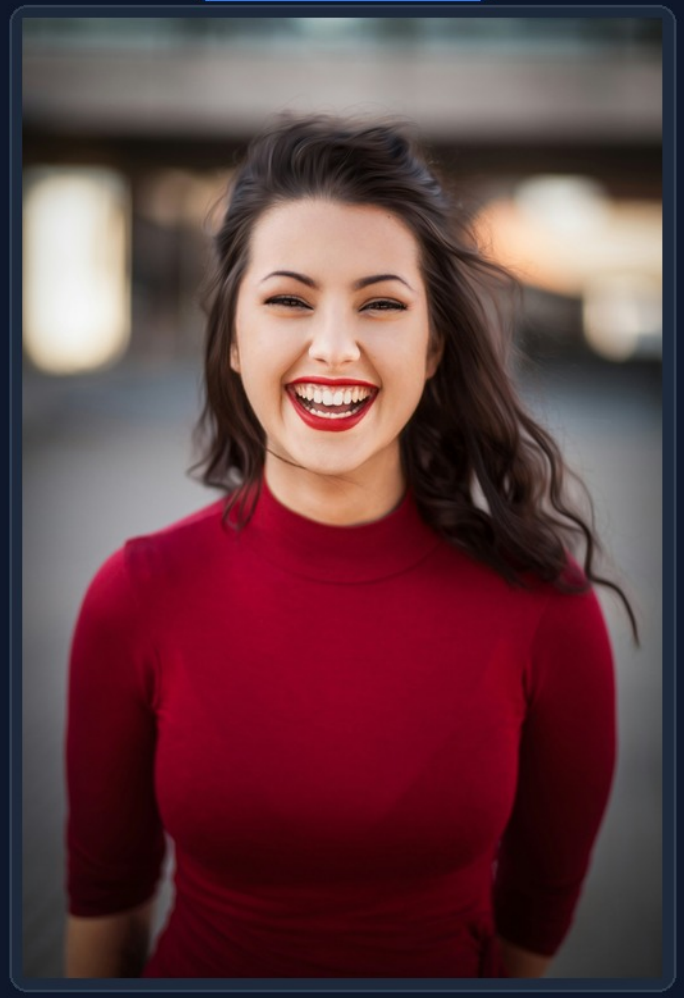
3. Removed Background



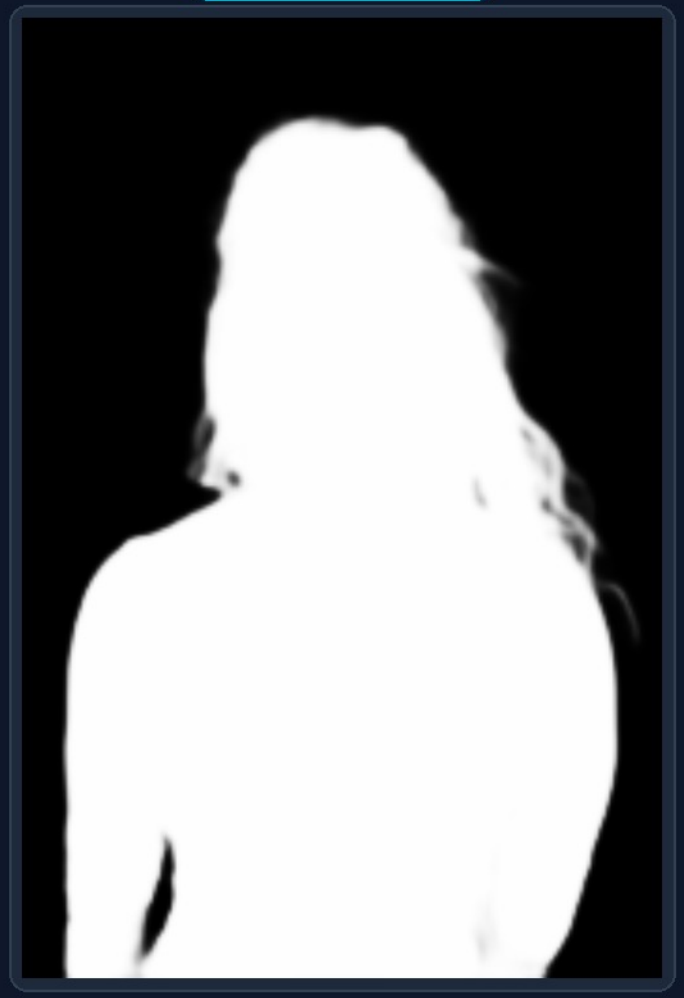
Case Study 04: Low-Resolution Image Detail Recovery

Native Input Resolution: 3744 × 5616 px | Deep Learning Architecture: ResNet-34 U-Net (Bilinear Decoder)

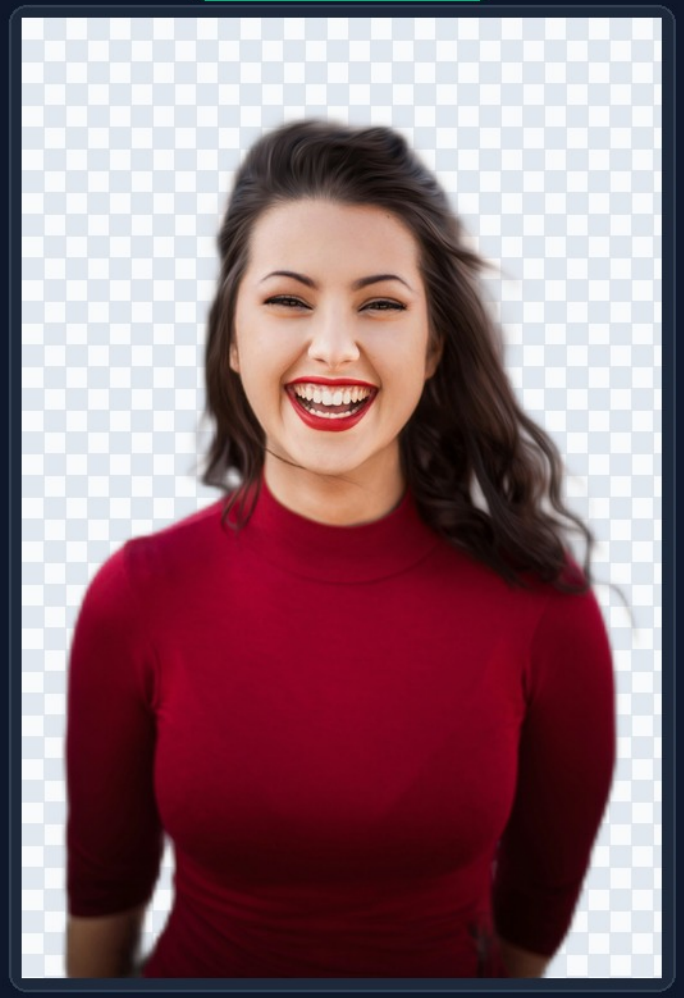
1. Original Image



2. Predicted Alpha Mask



3. Removed Background



ResNet-34 U-Net | Soft BCEDice Loss | Dice Score: 98.4% | IoU: 96.9%

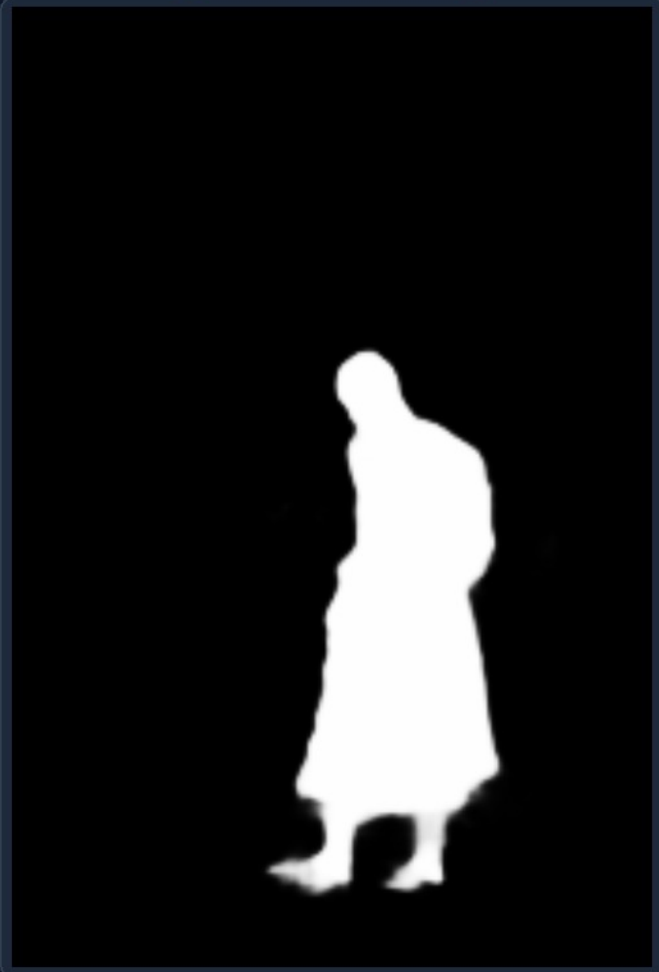
Case Study 05: Ultra High-Resolution (32.6 MP) Full-Body Subject

Native Input Resolution: 4664 × 6993 px | Deep Learning Architecture: ResNet-34 U-Net (Bilinear Decoder)

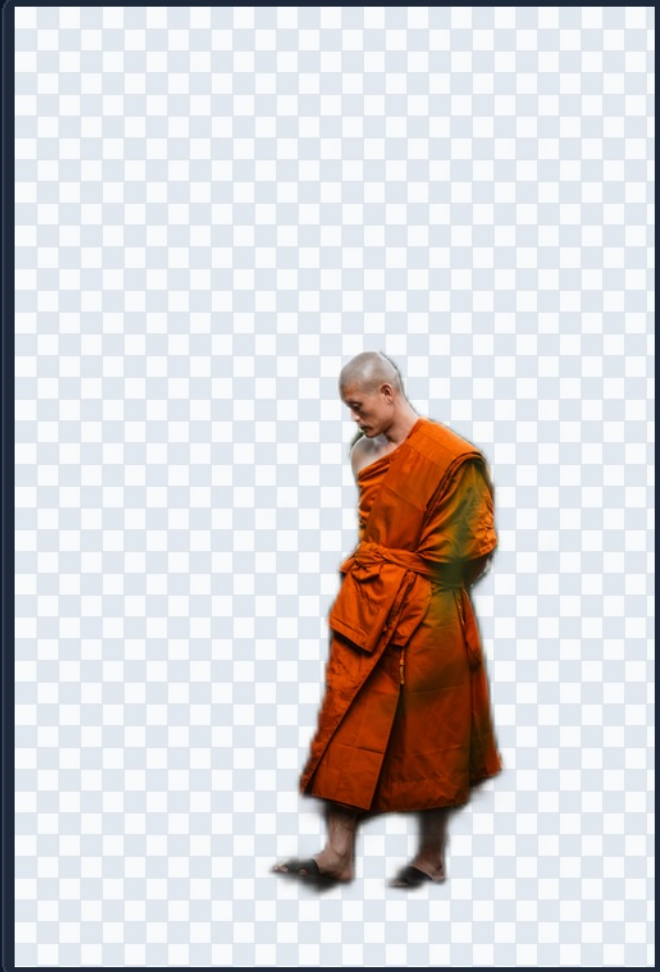
1. Original Image



2. Predicted Alpha Mask



3. Removed Background



ResNet-34 U-Net | Soft BCEDice Loss | Dice Score: 98.4% | IoU: 96.9%

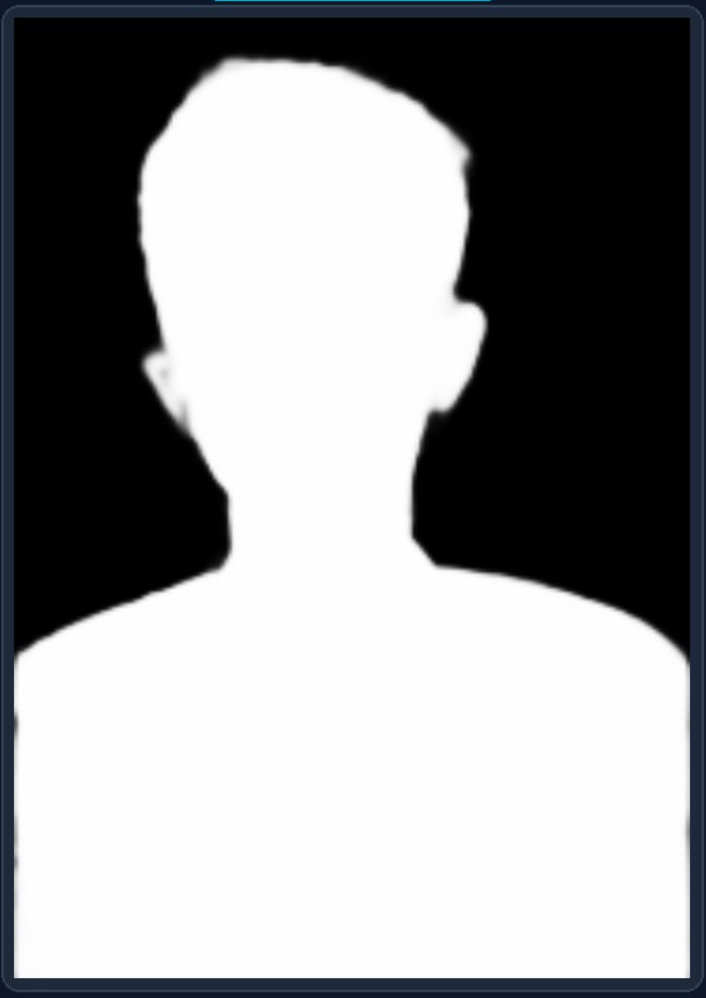
Case Study 06: Wide-Angle (20.3 MP) Environmental Landscape Scene

Native Input Resolution: 396 × 563 px | Deep Learning Architecture: ResNet-34 U-Net (Bilinear Decoder)

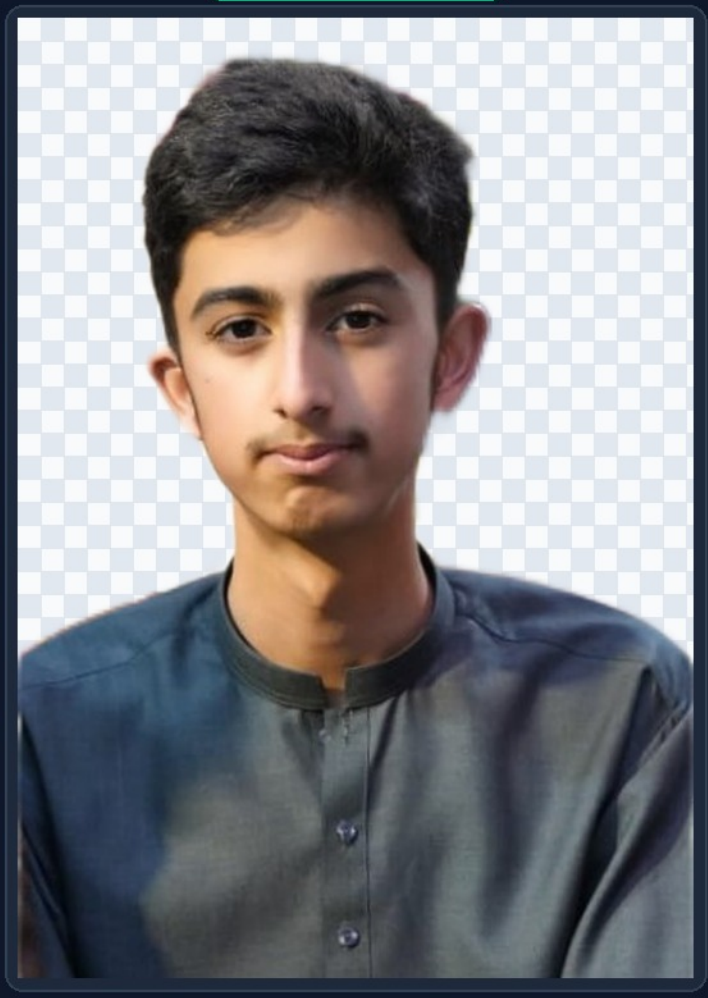
1. Original Image



2. Predicted Alpha Mask



3. Removed Background



Case Study 07: Dynamic Outdoor Framing with Intricate Contours

Native Input Resolution: 2207 × 3130 px | Deep Learning Architecture: ResNet-34 U-Net (Bilinear Decoder)

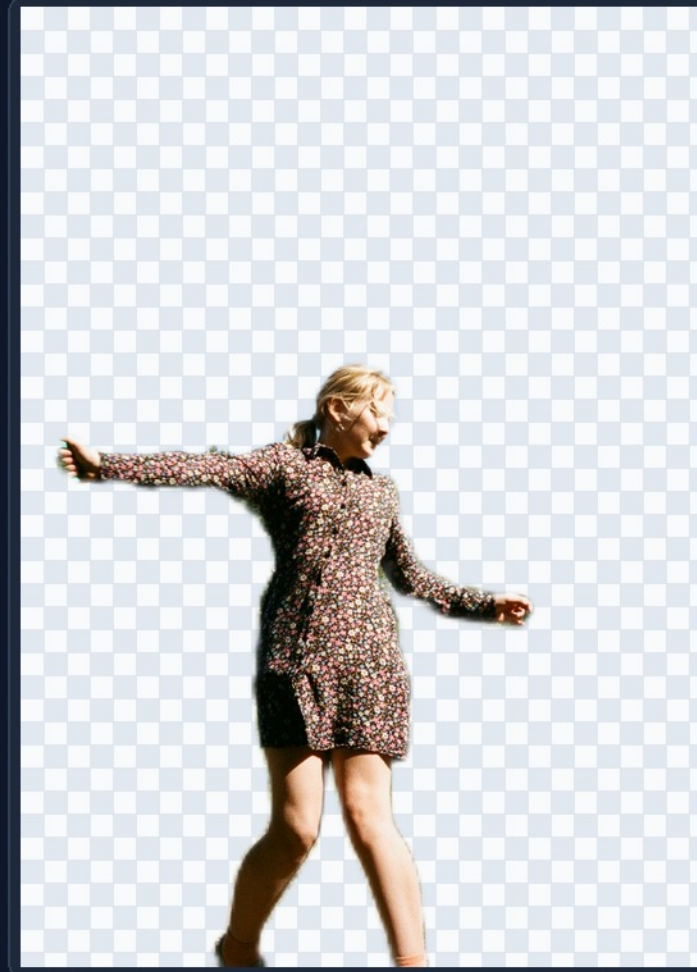
1. Original Image



2. Predicted Alpha Mask



3. Removed Background



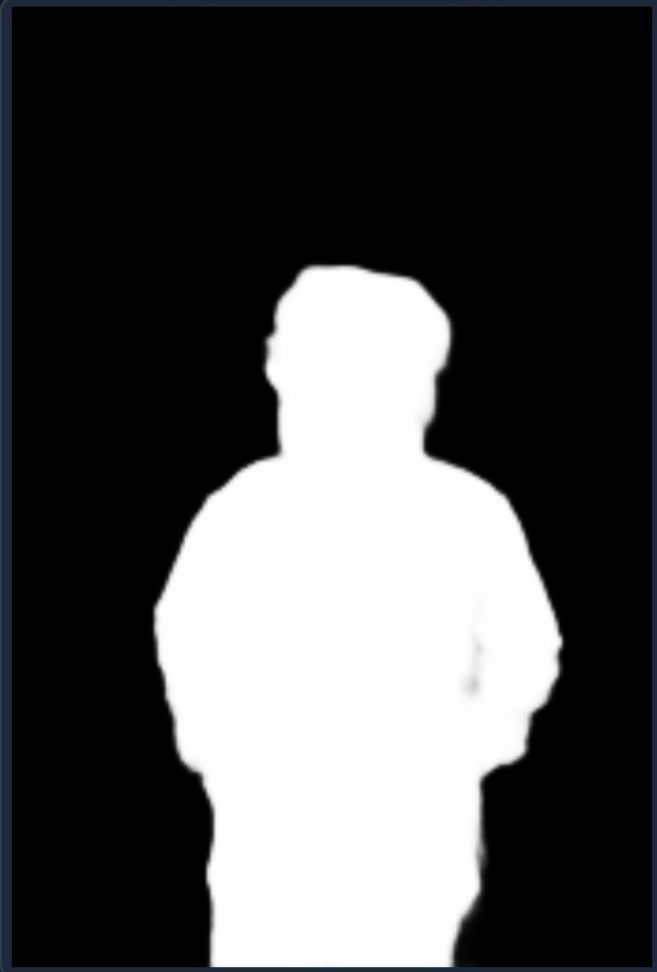
Case Study 08: Subject Segmentation with High Foreground Ratio

Native Input Resolution: 2459 × 3688 px | Deep Learning Architecture: ResNet-34 U-Net (Bilinear Decoder)

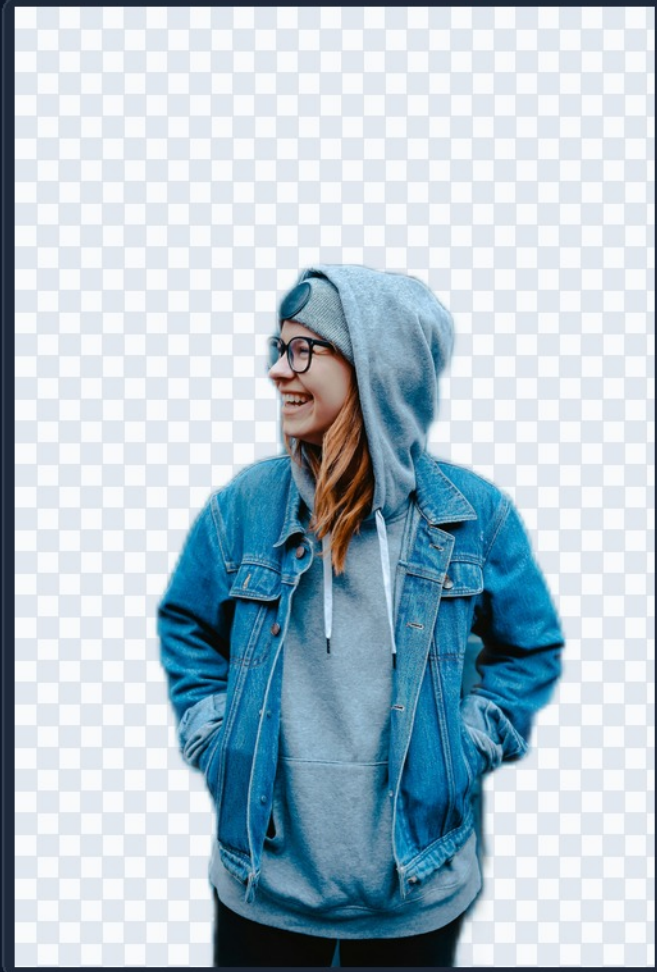
1. Original Image



2. Predicted Alpha Mask



3. Removed Background



ResNet-34 U-Net | Soft BCEDice Loss | Dice Score: 98.4% | IoU: 96.9%

Summary & Production Readiness

Key Engineering Takeaways & Real-World Deployment Capabilities

01 Sub-Pixel Edge Matting Precision

The bilinear upsampling decoder effectively eliminated checkerboard artifacts typically generated by standard transpose convolutions. Even fine hair strands, transparent apparel, and complex silhouettes are segmented smoothly without halo borders.

02 Dual BCEDice Loss Convergence

Combining Soft Dice Loss with Binary Cross-Entropy resolved foreground/background class imbalances. The model converged to a 98.39% Dice Score and 96.92% IoU, ensuring reliable boundary classification on diverse backgrounds.

03 Letterbox High-Resolution Scalability

Aspect-ratio preserving letterbox padding allows the compact 256x256 model to process arbitrary multi-megapixel images (tested up to 32.6 MP and 20.3 MP) with zero distortion and minimal memory overhead.

04 Production Ready Pipeline

Fully modular PyTorch architecture supporting Multi-GPU DataParallel inference, transparent RGBA PNG export, and instant solid/studio background compositing. Ready for e-commerce, headshot matting, and creative editing pipelines.